

1. (35 points) Evaluate the following integrals.

(a) $\int \frac{\sin(\ln x)}{x} dx$

(b) $\int_0^\pi (3x + 1) \cos(2x) dx$

(c) $\int \frac{2x^2 - 3x + 18}{x(3 - x)(2x + 3)} dx$

(d) $\int_0^{\pi/2} \sin^2(3x) \cos^3(3x) dx$

(e) $\int \frac{x + 2}{x^2 + 2x + 5} dx$

(f) $\int \arctan(\sqrt{x}) dx$

(g) $\int \frac{1}{x^2\sqrt{4 - x^2}} dx$

2. (8 points) Evaluate the following limits or explain why they do not exist.

(a) $\lim_{x \rightarrow 0^+} (1 + \sin(2x))^{\cot(x)}$

(b) $\lim_{x \rightarrow 0} \frac{e^{4x} - 4x - 1}{x^2}$

3. (4 points) Evaluate the improper integral or explain why it is divergent: $\int_0^\infty x^2 e^{-x^3} dx$

4. (5 points) Find the volume of the solid whose base is the region enclosed by $y = x^2 + x$ and $y = 0$, and whose cross sections perpendicular to the x -axis are squares.

5. (5 points) Solve for y given that

$$y^2 \frac{dy}{dx} \sec^2(x) = 1, \quad y(0) = 0.$$

6. (5 points) At 11AM, detectives discover a dead body in a pond and measure its temperature: 13°C . Two hours later, the body's temperature has dropped to 11°C . The pond's water temperature is constant at 10°C . We assume Newton's law of cooling: the rate of change of the temperature of the body is proportional to the temperature difference between the body and the pond.

- (a) Set up a differential equation with initial conditions that models the temperature of the body.
- (b) Solve the differential equation to find an expression for the temperature of the body as a function of time.

- (c) Determine the time of death, when the body temperature was a normal 37°C .

7. (5 points) Determine whether the sequence $\{a_n\}$ converges or diverges. Justify your answer: if the sequence converges, find the limit; otherwise, explain why it diverges.

(a) $a_n = \frac{\arctan(n)}{e^{1/n}}$

(b) $a_n = \frac{\ln(n^2)}{\ln(3n)}$

8. (13 points) Determine whether the following series are convergent or divergent. Justify your answers.

(a) $\sum_{n=3}^\infty \frac{n}{\sqrt{n^3 - 3n^2 + 2n}}$

(b) $\sum_{n=1}^\infty (-1)^n \left(\frac{n}{n+1}\right)^{n^2}$

(c) $\sum_{n=2}^\infty \frac{n! 7^n}{(2n-3)!}$

(d) $\sum_{n=1}^\infty \frac{\cos(n)}{1 + e^n}$

9. (6 points) Find the sum of each series:

(a) $\sum_{n=1}^\infty \frac{5}{n(n+1)}$

(b) $\sum_{n=1}^\infty \frac{2^n + 3}{4^n}$

10. (3 points) Using known series, find the Maclaurin series of

$$\frac{x^2}{1 - x^4}.$$

11. (5 points) (a) Using known series, find a power series representation of the function

$$\frac{1 - e^{-x^2}}{x^2}.$$

- (b) Use part (a) to find a power series representation of

$$\int \frac{1 - e^{-x^2}}{x^2} dx.$$

12. (6 points) Find the radius and the interval of convergence: $\sum_{n=1}^\infty \frac{(x+2)^n}{4^{n+1} \sqrt[3]{n}}$.

Answers

1. (a) $-\cos(\ln x) + C$

(b) 0

(c) $2 \ln |x| - \ln |3 - x| - 2 \ln |2x + 3| + C$

(d) $-\frac{2}{45}$

(e) $\frac{1}{2} \ln |(x+1)^2 + 4| + \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + C$

(f) $x \arctan(\sqrt{x}) - \sqrt{x} + \arctan(\sqrt{x}) + C$

(g) $-\frac{\sqrt{4-x^2}}{4x} + C$

2. (a) e^2

(b) 8

3. $\frac{1}{3}$

4. $\frac{1}{30}$

5. $y = \sqrt[3]{\frac{3x}{2} + \frac{3 \sin(2x)}{4}}$

6. (a) $\frac{dy}{dt} = k(y-10) \quad y(0) = 13 \quad y(2) = 11$

(b) $y(t) = 10 + 3 \left(\frac{1}{3}\right)^{t/2}$

(c) 7AM

7. (a) $\frac{\pi}{2}$

(b) 2

8. (a) D

(b) C

(c) C

(d) C

9. (a) 5

(b) 2

10. $\frac{x^2}{1-x^4} = \sum_{n=0}^{\infty} x^{4n+2}$

11. (a) $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{x^{2n-2}}{n!}$

(b) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n!} \cdot \frac{x^{2n-1}}{2n-1} + C$

12. $R = 4$ et I.C. = $[-6, 2)$